

RNA-seq Differential Expression — Colorectal Cancer

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1. Introduction and objectives

In this assignment I work with RNA-seq data from study GSE50760, available on GEO. The dataset contains expression data from colorectal-cancer patients, where each patient has three paired samples: normal colon, primary tumour, and liver metastasis.

What I want to do is identify which genes change their expression between normal tissue and primary tumour, using a paired approach since the samples come from the same patients. Then I look at which biological functions are enriched among the genes that go up in tumour, through an over-representation analysis on GO.

I used limma-voom for the differential-expression part because it handles the paired design well and it is what we saw in class.

2. Description of the data

2.1 The SummarizedExperiment object

The data are stored in an `se` object of class `SummarizedExperiment`. This is what it contains:

- 39,376 genes and 54 samples.
- 4 assays: counts, logCPM, TPM, FPKM. For the DE analysis I use the raw counts.
- In `colData` there are several variables, but the important ones are `patient_id` (to identify each patient) and `condition` (normal, primary_tumor, or liver_metastasis). Also present are `geo_accession`, `sample_type`, `pair_group`, etc.
- `rowData` holds the gene annotation: `gene_id`, `gene_symbol`, `entrez_id`, `has_entrez`, `description`, `chromosome`, and others.
- The design is paired: 18 patients, each with exactly 3 samples (one per condition), i.e. 54 in total.

Condition	Samples
normal	18
primary_tumor	18
liver_metastasis	18

Table 1. Sample distribution by condition in the full dataset.

3. Individual sample subset

I used seed 2943 to select my subset of 10 patients.

Selected patients: AMC_10, AMC_13, AMC_18, AMC_2, AMC_20, AMC_21, AMC_22, AMC_24, AMC_7, AMC_9. This gives me 30 samples (10 patients × 3 conditions each).

Table 2. Verification of the paired design in the subset: each patient has one sample per condition (all cells equal to 1 across the three conditions for the ten patients).

As the table shows, each patient has one sample per condition, so the pairing is correct.

4. Exploratory analysis and quality control

Before entering the DE analysis, I explored the full 30-sample subset to check for any obvious problems.

4.1 Boxplot of logCPM distributions

[Figure 1. Distribution of logCPM values across the 30 samples; colours indicate tissue condition.]

The distributions look fairly similar across all samples. The medians are at roughly the same level and no sample stands out as clearly different. There is no need to remove any.

4.2 PCA analysis

[Figure 2. PCA of the 30 samples based on the 1,000 most variable genes.]

In the PCA, PC1 (45.4% of the variance) clearly separates the normal samples from the tumoural ones. The normal-colon samples group together on the left. Primary tumours and metastases are on the right but overlap considerably, which makes sense because metastases arise from the primary tumour.

You can also see there is between-patient variability (the points within each condition are spread out), which is another reason we need the paired model — to control for those individual differences.

I see no sample that looks like a clear outlier, so I keep them all.

5. Preprocessing: filtering and normalisation

5.1 Filtering

From here I work only with the normal and primary_tumor samples (20 samples, 10 per group).

- Genes before filtering: **39,376**
- Genes after filtering: **18,961**
- Removed: 20,415 genes

I used `filterByExpr()` from `edgeR`, which sets a minimum count threshold automatically based on library sizes and the smallest group. It essentially removes genes with counts too low to be tested reliably. More than half the genes are removed, which is normal in RNA-seq — many genes are simply not expressed in these tissues.

5.2 Normalisation

I applied TMM normalisation. TMM computes a scaling factor for each sample to compensate for differences in RNA composition between libraries. It is the default and recommended normalisation for `edgeR`/`limma-voom` workflows. The idea is that without normalisation, a few highly expressed genes in one sample could make the other genes appear artificially lower.

The normalisation factors I obtained range from 0.836 to 1.165, all fairly close to 1, so there are no serious composition problems.

6. Experimental design and model

6.1 Why a paired model is needed

Since each patient contributes both a normal and a tumour sample, the two samples from the same patient are not independent — they share the same genetic background, age, lifestyle, etc. If we simply compared all normals against all tumours ignoring which patient they came from, the between-patient variability would add noise and make it harder to detect real differences. The paired design controls for this by estimating the tumour effect within each patient.

6.2 Design matrix

The formula I used is:

```
~ patient_id + condition
```

`patient_id` accounts for the differences between patients, and `condition` is what we are really interested in — it gives the tumour-vs-normal comparison. The reference level for `condition` is “normal”, so the `conditionprimary_tumor` coefficient represents the log fold-change of tumour relative to normal.

The design matrix ends up with 11 columns: 1 intercept + 9 patient effects + 1 condition effect.

7. Differential-expression analysis

I used limma-voom. Voom estimates the mean-variance trend from the counts and assigns precision weights to each observation, then limma fits a linear model per gene and uses empirical Bayes to share information across genes in the variance estimates.

7.1 Summary of results

Using $FDR < 0.05$ and $|\log_2FC| > 2$ as thresholds:

- Total DEGs: **1,010**
- Up-regulated in tumour: **280**
- Down-regulated in tumour: **730**

There are many more genes going down in tumour than up. This is probably because the tumour loses much of the normal differentiation programme of the colon epithelium.

7.2 Top 10 differentially expressed genes

Gene	logFC	MeanExpr	P-value	FDR
BEST4	-7.07	3.53	2.00e-11	1.90e-07
OTOP3	-7.01	-2.02	1.96e-11	1.90e-07
OTOP2	-8.48	1.68	3.05e-11	1.93e-07
CA7	-6.39	3.61	9.72e-10	4.61e-06
ABCG2	-4.25	2.85	3.64e-09	1.38e-05
SPIB	-5.25	3.90	1.53e-08	4.82e-05
MTHFD1L	2.11	3.84	2.12e-08	5.74e-05
SLC17A8	-5.03	-1.78	2.64e-08	6.25e-05
PHLPP2	-2.10	5.41	4.22e-08	8.01e-05
UNC5C	-2.12	2.69	6.44e-08	9.40e-05

Table 3. Top 10 DEGs ordered by adjusted p-value.

Most of the top hits are down-regulated. BEST4 has the lowest FDR and a logFC of about -7, meaning it practically disappears in tumour tissue. BEST4 is a marker of a specific type of intestinal epithelial cell, so it makes sense that it vanishes when the tissue turns cancerous. OTOP2 and OTOP3 are also strongly down-regulated. The only clearly up-regulated gene in the top 10 is MTHFD1L, involved in one-carbon/folate metabolism — cancer cells tend to up-regulate these pathways because they need them to proliferate quickly.

7.3 Volcano plot

[Figure 3. Volcano plot. Red = up-regulated in tumour, blue = down-regulated, grey = not significant. Dashed lines mark the thresholds used.]

The asymmetry is clear — the left side (down-regulated) has more coloured points and they extend further than the right side (up-regulated).

8. Gene annotation

The annotation was already in `rowData(se)` with `gene_symbol`, `entrez_id`, and a boolean `has_entrez`.

- DEGs with valid Entrez ID: **1,010 / 1,010**
- Up-regulated with Entrez ID: **280 / 280**
- Down-regulated with Entrez ID: **730 / 730**

I used Entrez IDs for the ORA since that is what `org.Hs.eg.db` expects. All DEGs turned out to have valid Entrez IDs, so no gene had to be removed for lack of annotation.

9. Over-representation analysis (ORA)

For the ORA I used a hypergeometric test (Fisher's exact test) on GO Biological Process terms. I used only the genes up-regulated in tumour, as the assignment asks. The universe is all genes that passed filtering.

9.1 ORA results

- Up-regulated genes used: 280
- GO-BP terms tested: 436 (only terms with 10–500 genes in the universe and at least 2 overlapping DEGs)
- Significant terms after BH correction (FDR < 0.05): 127

9.2 Most enriched GO-BP terms

GO ID	Term	Gene Ratio	FDR	Count
GO:0006935	chemotaxis	16/280	4.94e-08	16
GO:0006954	inflammatory response	26/280	5.85e-08	26
GO:0030593	neutrophil chemotaxis	10/280	1.95e-07	10
GO:0007267	cell-cell signaling	16/280	2.11e-07	16
GO:0030199	collagen fibril organization	11/280	2.11e-07	11
GO:0006508	proteolysis	26/280	5.17e-07	26
GO:0006952	defense response	11/280	4.16e-06	11
GO:0061844	antimicrobial humoral response (antimicrobial peptide)	10/280	4.16e-06	10
GO:0030198	extracellular matrix organization	12/280	1.26e-05	12
GO:0006955	immune response	22/280	3.19e-05	22

Table 4. Top 10 GO-BP terms enriched among the up-regulated genes.

9.3 Dotplot of enriched terms

[Figure 4. Top 15 enriched GO-BP terms. Point size = number of genes, colour = FDR.]

9.4 Biological interpretation

The most enriched term is chemotaxis, followed by inflammatory response and neutrophil chemotaxis. Looking at the full list, the enriched terms are mostly related to inflammation (chemotaxis, neutrophil chemotaxis, inflammatory response), extracellular-matrix remodelling (ECM organization, collagen fibril organization), and immune signalling (defense response, immune response).

This makes sense for colorectal cancer. Tumours are known to have heavy infiltration of immune cells, especially neutrophils, which explains the chemotaxis and inflammation terms. The ECM and collagen terms are also expected — tumours need to degrade and remodel the surrounding tissue to grow and eventually spread. A term for positive regulation of the epithelial-mesenchymal transition (EMT) also appears, which is one of the key steps in how cancer cells become more invasive and mobile.

Basically the genes going up in tumour are doing things related to inflammation, tissue remodelling, and invasion, which is consistent with what is known about colorectal-cancer biology.

10. Discussion and conclusions

I found 1,010 DEGs between primary tumour and normal colon (FDR < 0.05, $|\log_2\text{FC}| > 2$). Most (730) are down-regulated in tumour, and only 280 go up. The down-regulation pattern probably reflects the loss of the colon's normal differentiation programmes. The up-regulated genes are enriched in

immune/inflammatory processes and ECM remodelling, which is what you would expect from a growing tumour.

A few things to keep in mind:

- Using the paired model was important. The PCA already showed patient-level variability, so ignoring the pairing would have added noise and probably given us fewer DEGs.
- Filtering removed about half the genes, which is typical. TMM normalisation is standard for this kind of workflow.
- I only ran the ORA on up-regulated genes as the assignment asked. It would also be interesting to see what the down-regulated genes are enriched for — probably things like ion transport, absorption, and other normal colon functions.
- I used a manual Fisher exact test instead of clusterProfiler because I could not install it in my environment. The underlying test is the same (hypergeometric), so the results should be equivalent.
- Working with 10 of the 18 patients is a limitation, but the paired design helps compensate for the smaller sample by reducing noise.

In summary, the analysis shows that primary colorectal tumours switch on genes involved in inflammation, matrix remodelling, and EMT, while switching off many normal epithelial functions. These findings agree with what has been reported in the literature on this type of cancer.

11. Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE, fig.align = "center")

library(SummarizedExperiment)
library(edgeR)
library(limma)
library(org.Hs.eg.db)
library(GO.db)
library(AnnotationDbi)
library(ggplot2)
library(pheatmap)
library(ggrepel)

load("PEC2_GSE50760_colorectal_SE.Rda")

# sample distribution by condition (full dataset)
knitr::kable(as.data.frame(table(se$condition)),
             col.names = c("Condition", "Samples"))

# personal subset of 10 patients
set.seed(2943)
patients <- unique(se$patient_id)
selected_patients <- sample(patients, 10)
se_sub <- se[, se$patient_id %in% selected_patients]

# check the paired design in the subset
knitr::kable(table(se_sub$patient_id, se_sub$condition))

# logcpm values for exploration
dge_exp <- DGEList(counts = assay(se_sub, "counts"))
dge_exp <- calcNormFactors(dge_exp, method = "TMM")
logcpm <- cpm(dge_exp, log = TRUE, prior.count = 2)

# prepare data for the boxplot
sample_info <- data.frame(
  sample = colnames(logcpm),
  condition = as.character(se_sub$condition),
  patient = as.character(se_sub$patient_id),
  stringsAsFactors = FALSE
)

# short label
sample_info$label <- paste0(substr(sample_info$condition, 1, 3), "_", sample_info$patient)

# long format for ggplot
box_data <- data.frame(
  value = as.vector(logcpm),
```

```

    sample = rep(colnames(logcpm), each = nrow(logcpm)),
    stringsAsFactors = FALSE
  )
  box_data$condition <- sample_info$condition[match(box_data$sample, sample_info$sample)]
  box_data$label <- sample_info$label[match(box_data$sample, sample_info$sample)]

  ggplot(box_data, aes(x = reorder(label, as.numeric(factor(condition))), y = value, fill =
condition)) +
    geom_boxplot(outlier.size = 0.3, outlier.alpha = 0.3) +
    theme_bw(base_size = 10) +
    theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 6)) +
    labs(x = "", y = "logCPM", fill = "Condition") +
    scale_fill_manual(values = c("normal" = "#4DAF4A", "primary_tumor" = "#E41A1C",
"liver_metastasis" = "#377EB8"))

  # most variable genes for the pca
  vars <- apply(logcpm, 1, var)
  top_genes <- names(sort(vars, decreasing = TRUE))[1:1000]
  pca <- prcomp(t(logcpm[top_genes, ]), scale. = TRUE)
  pca_df <- data.frame(
    PC1 = pca$x[, 1], PC2 = pca$x[, 2],
    condition = as.character(se_sub$condition),
    patient = as.character(se_sub$patient_id)
  )
  var_explained <- round(summary(pca)$importance[2, 1:2] * 100, 1)

  ggplot(pca_df, aes(x = PC1, y = PC2, color = condition)) +
    geom_point(size = 3, alpha = 0.8) +
    theme_bw(base_size = 11) +
    labs(x = paste0("PC1 (", var_explained[1], "%)"),
    y = paste0("PC2 (", var_explained[2], "%)"), color = "Condition") +
    scale_color_manual(values = c("normal" = "#4DAF4A", "primary_tumor" = "#E41A1C",
"liver_metastasis" = "#377EB8"))

  # keep only normal and primary_tumor
  se_pn <- se_sub[, se_sub$condition %in% c("normal", "primary_tumor")]
  se_pn$condition <- droplevels(se_pn$condition)
  se_pn$patient_id <- droplevels(factor(se_pn$patient_id))

  dge <- DGEList(counts = assay(se_pn, "counts"), genes = as.data.frame(rowData(se_pn)))
  genes_before <- nrow(dge)
  keep <- filterByExpr(dge, group = se_pn$condition)
  dge <- dge[keep, , keep.lib.sizes = FALSE]
  genes_after <- nrow(dge)
  dge <- calcNormFactors(dge, method = "TMM")

  # paired design matrix
  design <- model.matrix(~ patient_id + condition, data = data.frame(
    patient_id = se_pn$patient_id,
    condition = relevel(se_pn$condition, ref = "normal")
  ))

  # limma-voom
  v <- voom(dge, design, plot = FALSE)
  fit <- lmFit(v, design)
  fit <- eBayes(fit)

  results <- topTable(fit, coef = "conditionprimary_tumor", number = Inf, sort.by = "none")
  degs <- results[results$adj.P.Val < 0.05 & abs(results$logFC) > 2, ]
  up <- degs[degs$logFC > 0, ]
  down <- degs[degs$logFC < 0, ]

  # top 10 table
  top10 <- head(degs[order(degs$adj.P.Val), ], 10)

  # volcano plot categories
  results$significance <- "Not significant"
  results$significance[results$adj.P.Val < 0.05 & results$logFC > 2] <- "Up in tumour"
  results$significance[results$adj.P.Val < 0.05 & results$logFC < -2] <- "Down in tumour"
  results$neg_log10_fdr <- -log10(results$adj.P.Val)
  results$neg_log10_fdr[results$neg_log10_fdr > 15] <- 15 # cap for visualisation

  # over-representation analysis (manual hypergeometric test)
  up_annotated <- up[!is.na(up$entrez_id) & up$has_entrez == TRUE, ]
  up_entrez <- as.character(up_annotated$entrez_id)

```

```

TRUE]) universe <- as.character(results$entrez_id[!is.na(results$entrez_id) & results$has_entrez ==
# GO-BP annotations
go_bp_all <- AnnotationDbi::select(org.Hs.eg.db,
  keys = keys(org.Hs.eg.db, "ENTREZID"),
  columns = c("GO", "ONTOLOGY"), keytype = "ENTREZID")
go_bp_all <- go_bp_all[go_bp_all$ONTOLOGY == "BP" & !is.na(go_bp_all$GO), ]

# map GO term -> genes, filter by term size (10-500 genes in the universe)
go2gene <- split(go_bp_all$ENTREZID, go_bp_all$GO)
term_sizes <- sapply(go2gene, function(x) length(intersect(unique(x), universe)))
go2gene_filtered <- go2gene[term_sizes >= 10 & term_sizes <= 500]

# fisher/hypergeometric test per term
N <- length(universe)
k <- length(intersect(up_entrez, universe))
ora_list <- list()
for (go_id in names(go2gene_filtered)) {
  go_genes <- unique(go2gene_filtered[[go_id]])
  go_in_univ <- intersect(go_genes, universe)
  m <- length(go_in_univ)
  overlap <- intersect(up_entrez, go_in_univ)
  q <- length(overlap)
  if (q < 2) next
  pval <- phyper(q - 1, m, N - m, k, lower.tail = FALSE)
  ora_list[[go_id]] <- data.frame(GOID = go_id, GeneRatio = paste0(q, "/", k),
    BgRatio = paste0(m, "/", N), pvalue = pval, Count = q)
}
ora_results <- do.call(rbind, ora_list)
ora_results$p.adjust <- p.adjust(ora_results$pvalue, method = "BH")

# add GO term names
go_terms <- AnnotationDbi::select(GO.db, keys = ora_results$GOID, columns = "TERM", keytype =
"GOID")
ora_results$Term <- go_terms$TERM[match(ora_results$GOID, go_terms$GOID)]
ora_sig <- ora_results[ora_results$p.adjust < 0.05, ]
ora_sig <- ora_sig[order(ora_sig$p.adjust), ]

# dotplot of the top 15 terms
ora_plot <- head(ora_sig, 15)
ora_plot$Term <- factor(ora_plot$Term, levels = rev(ora_plot$Term))
ggplot(ora_plot, aes(x = Count, y = Term, size = Count, color = p.adjust)) +
  geom_point() +
  scale_color_gradient(low = "red", high = "blue", name = "FDR") +
  theme_bw(base_size = 10) +
  labs(x = "Number of genes", y = "", size = "Count")

```